

SANJAY DESHMUKH

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EDUCATION

Bachelor of Technology – Vishwakarma Institute of Technology (VIT), Pune

Aug 2023 - May 2027

- Current CGPA : 8.86

SKILLS

- **Languages:** C, C++, Java, Python, R, JavaScript
- **Web Development:** HTML, CSS, ReactJS, Spring Boot, Flask, TypeScript, Express.js, Node.js
- **Databases:** MySQL, PostgreSQL, MongoDB, Neo4J, Chroma, Qdrant, Pinecone
- **Tools & Devops:** Docker, Kubernetes, AWS, Git, GitHub, TensorRT, PyTorch, TensorFlow, REST, SOAP APIs, n8n, CI/CD
- **AI/Generative AI:** LangChain, Ollama, OpenAI, APIs, Prompt Engineering, Large Language Models (LLMs), Retrieval-Augmented Generation(RAG), Vector Databases, Agentic AI, OpenCV

PROJECTS

1. Wheat Grain Quality Classification System

- Built an AI-powered web application using Flask + ResNet50 for automated wheat grain quality grading.
- Extracted 2048-D deep features and classified grains into A–F quality grades using ML models.
- Developed automated PDF reporting with grade distribution, image analysis, and performance statistics.
- Published a custom wheat dataset on Kaggle and Hugging Face for agricultural AI research.

2. Candlelytics – Demand Forecasting Dashboard

- Built an AI-powered forecasting system using Prophet, LSTM, and XGBoost for candle demand prediction.
- Developed an ensemble time-series model to improve forecast accuracy.
- Enabled next-month and long-term production planning with downloadable forecast reports.

3. AirCursor – Hand Gesture Controlled Mouse

- Built a webcam-based virtual mouse using MediaPipe hand tracking.
- Enabled cursor movement, clicking, and scrolling using finger gestures.
- Implemented real-time gesture recognition with OpenCV and PyAutoGUI.

4. VoteGuard – Voting Anomaly Detection using Reinforcement Learning

- Built a real-time anomaly detection dashboard using Flask and reinforcement learning-based online learning.
- Detected voting irregularities using ensemble ML models with an adaptive RL agent.
- Designed an interactive dashboard with live charts, alerts, and anomaly logs.

EXTRA-CURRICULAR ACTIVITIES

- Published 3 Research papers (2 in IEEE and 1 in Taylor and Francis)
- Got the Best Paper Award at the IEEE International Conference on Data, Energy and Communication Networks (IC-DECoN 2025), for my paper on "UAV-Assisted AI-Powered Crack Detection in High-Rise Buildings"

LEADERSHIP

- Secretary of ICON Club of VIT Pune.
- AI/ML Head of International Society of Automation, Pune section
- Research Analyst at The Investment Forum, VIT Pune.

INTERNSHIPS AND CERTIFICATIONS

- JPMorgan Chase – Software Engineering Job Simulation (Virtual) ([LINK](#))
- Deloitte – Data Analytics Job Simulation (Virtual) ([LINK](#))
- AWS – Cloud Technology Consultant Specialization ([LINK](#))
- Google – Project Management Specialization ([LINK](#))